Ancilla-mediated fixed-point quantum search using Grover iterations

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Grover's quantum search algorithm provides a fundamental quadratic speedup for unstructured datasets, reducing query complexity from $\mathcal{O}(N)$ to $\mathcal{O}(\sqrt{N})$. However, the algorithm's reliance on precise iteration counts leads to the "soufflé problem," where over-rotation results in a sharp decline in success probability. This limitation is particularly restrictive when the number of solution states, M, is unknown. In this work, we present an ancilla-mediated fixed-point quantum search algorithm that achieves robust convergence by mapping the solution amplitude to a dedicated ancilla qubit. Unlike existing phase-matching fixed-point methods, our approach utilizes Grover's real-plane reflections, thereby maintaining the intuitive geometric architecture of the original algorithm. We demonstrate that this method achieves a success probability of at least 92.6% with a query complexity of approximately $\mathcal{O}(\sqrt{N/M})$, effectively bridging the gap between standard amplitude amplification and robust fixed-point convergence.

I. INTRODUCTION

Quantum computing leverages phenomena such as superposition and entanglement to solve problems that remain computationally expensive for classical architectures [1–7]. A cornerstone of this field is Grover's search algorithm, which offers a quadratic speedup for searching unstructured datasets [7–10]. For a dataset of N entries containing M solutions, Grover's algorithm identifies a target state in $\mathcal{O}(\sqrt{N/M})$ iterations, whereas a classical linear search requires $\mathcal{O}(N/M)$ queries [7, 8, 11–13].

Despite its efficiency, the standard Grover algorithm requires precise control over the number of iterations to avoid the "soufflé problem" [14]. Because the success probability follows a sinusoidal curve, over-rotation or under-rotation leads to the selection of non-solution states. This sensitivity is particularly problematic in real-world applications where the exact number of solution states, M, is unknown. To address this lack of robustness, various fixed-point quantum search strategies have been proposed to replace Grover's selective inversions with monotonic convergence mechanisms [14–16].

Grover's initial fixed-point variant utilised $\pi/3$ phase shifts to create attraction basins, ensuring convergence regardless of iteration counts, though it maintained a standard $\mathcal{O}(\sqrt{N})$ scaling [14]. Subsequent refinements by Yoder *et al.* [16] and Li *et al.* [17] established rigorous convergence bounds via a phase-matching approach; in particular, Li *et al.* achieve a query complexity $\mathcal{O}(5.643\sqrt{N/M})$ for an unknown number of solutions, giving a quadratic speedup.

Alternative methodologies for handling unknown M include various trial-and-error techniques [9, 18]. The foundational approach proposed by Boyer *et al.* [9] involves executing Grover's algorithm across multiple trials, with iteration counts chosen randomly from an exponentially increasing interval. This stochastic framework ensures that the average success probability remains at least $1/4$ once the algorithm reaches a critical stage, enabling the discovery of a target item with an expected query complexity of at most $4\sqrt{N/M}$. While effective, this approach is classified as a randomized application of Grover's algorithm [17–19].

Closest to the present work are two earlier *measurement-based* ancilla schemes. Patel *et al.* [20] introduced a recursive measurement-driven construction in which ancilla-controlled selective inversions and diffusion operators drive the register monotonically toward the target; for a single marked item, however, the recursion requires $\mathcal{O}(N)$ queries. Mani and Patvardhan [21] subsequently recovered the canonical $\mathcal{O}(\sqrt{N})$ scaling with a fast measurement-based scheme that thresholds ancilla-measurement outcomes as a stopping rule, at the cost of roughly twice the oracle calls per iteration.

Our protocol shares the ancilla-measurement philosophy of these works but differs in its construction and guarantees. We retain Grover's real-plane reflections, use no phase gates, and employ a single ancilla solely as a measurable success flag within a deterministic geometric doubling schedule. This yields a provable success probability of at least 92.6% with an expected query complexity of $\mathcal{O}(\sqrt{N/M})$, where M is unknown. In the worst case where no solution exists ($M = 0$), the algorithm exhausts the schedule and terminates after $\mathcal{O}(\sqrt{N})$ oracle calls, dominated by the final cumulative count $2^{r-1} \sim \frac{\pi}{2}\sqrt{N}$.

The paper is organized as follows. Section II provides an introduction to the quantum search algorithms. Section III provides the probability calculations and an analysis of the query complexity. Section IV compares the

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existing algorithms with the algorithm in this paper and provides a brief conclusion.

II. PRELIMINARIES AND FRAMEWORK

This work combines the amplitude amplification algorithm [7, 8] with a novel approach to the fixed-point algorithm to confidently reach the solution state, thereby successfully completing the search. The said algorithms are given below.

A. Amplitude amplification algorithm

The amplitude amplification algorithm is based on the Grover search algorithm, which is a quantum algorithm designed for unstructured search problems [7]. The quantum Oracle $\hat{O}$ is used to identify the solution state $|S\rangle$ from the set of all possible dataset states with high probability [7, 8, 11–13]. The Oracle $\hat{O}$ marks the solution state(s) by flipping their phase. Acting on a computational-basis state $|x\rangle$, it is the unitary

$$\hat{O}|x\rangle = \begin{cases} -|x\rangle, & x \text{ is a solution,} \\ |x\rangle, & \text{otherwise,} \end{cases} \qquad \hat{O} = \hat{I} - 2\sum_{s\in\mathcal{S}} |s\rangle\langle s|, \tag{1}$$

where $\mathcal{S}$ is the set of solutions; equivalently, $\hat{O}|S\rangle = -|S\rangle$ and $\hat{O}|R\rangle = |R\rangle$ for the collective solution and non-solution states $|S\rangle$ and $|R\rangle$. In this work, the dataset is encoded in n data qubits $q_1, \cdots, q_n$ in initial state $|\psi\rangle$, using Basis/Binary encoding [22, 23] by a state preparation operator $\hat{P}$ for some θ as

$$\hat{P}|0\rangle^{\otimes n} = |\psi\rangle = \cos\theta|R\rangle + \sin\theta|S\rangle. \tag{2}$$

The state of the ancilla qubit a is shown by $|\psi_a\rangle$ with the initial state taken to be $|0_a\rangle$. The combined initial state of data qubits and ancilla is represented by $|0_a\rangle|\psi\rangle$.

Amplitude amplification operator

The amplitude amplification operator ($\hat{G}$) can be defined to amplify the amplitude of the solution state $|S\rangle$. This is achieved by using Grover's algorithm [7]. The whole action can be defined as

$$\begin{aligned} \hat{G}|\psi\rangle &= \hat{G}(\cos\theta|R\rangle + \sin\theta|S\rangle) \\ &= \cos(3\theta)|R\rangle + \sin(3\theta)|S\rangle. \end{aligned} \tag{3}$$

Here, an operation of $\hat{G}$ has increased the solution state amplitude from $\sin\theta$ to $\sin(\theta + 2\theta)$. The quantum circuit for the operation can be given by:

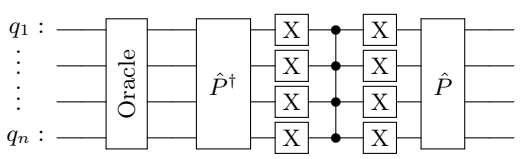


FIG. 1. Amplitude amplification operator $\hat{G}$ acting on the data register $q_1, \ldots, q_n$ (no ancilla is used here). "Oracle" and $\hat{P}$ denote the circuits for the oracle $\hat{O}$ and the state-preparation operator $\hat{P}$ of Eq. (2).

There is an increment of angle 2θ by each operation of $\hat{G}$. After k iterations of amplitude amplification $\hat{G}^k$ the state changes to

$$\hat{G}^k|\psi\rangle = \cos(\theta + 2k\theta)|R\rangle + \sin(\theta + 2k\theta)|S\rangle. \tag{4}$$

Here, the solution state amplitude increased from $\sin\theta$ to $\sin(\theta + 2k\theta)$. For a known number of solution states M, θ is given by

$$\theta = \arcsin\left(\sqrt{\frac{M}{N}}\right). \tag{5}$$

The solution state $|S\rangle$ attains its maximum amplitude when $\theta + 2k\theta = \frac{\pi}{2}$. This gives the optimal number of amplitude-amplification iterations ϵ as

$$\epsilon = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{M}} - \frac{1}{2} \right\rfloor, \tag{6}$$

where $\lfloor\cdot\rfloor$ denotes the floor function and θ is small.

Next, we look at the fixed-point search method.

B. Fixed point search

The fixed-point quantum search is a method that fixes a point of convergence, reaching which the search is deemed complete. In the proposed algorithm, an ancilla qubit in state $|1\rangle$ acts as a point of convergence for its respective data qubit [14, 16, 24, 25]. For an $n+1$ qubit system, with n data qubits $q_1, q_2, \cdots, q_n$ and an ancilla a, we define the state containing M solutions as $|S\rangle$:

$$|S\rangle = \frac{s_1|S_1\rangle}{|S|} + \frac{s_2|S_2\rangle}{|S|} + \cdots + \frac{s_M|S_M\rangle}{|S|}, \tag{7}$$

where $|S|$ is the normalising factor for the state $|S\rangle$ and s_i is the coefficient of the state $|S_i\rangle$ for all $i \in \{1, 2, \cdots, M\}$. Similarly, we write the rest of the $N - M$ non-solutions as $|R\rangle$:

$$|R\rangle = \frac{r_1|R_1\rangle}{|R|} + \frac{r_2|R_2\rangle}{|R|} + \cdots + \frac{r_{N-M}|R_{N-M}\rangle}{|R|}, \tag{8}$$

where $|R|$ is the normalising factor for the state $|R\rangle$ and r_i is the coefficient of the state $|R_i\rangle$ for all $i \in \{1, 2, \cdots, N-M\}$.

The initial state $|\Psi\rangle$ for the n data qubits and ancilla a for the above basis is given as

$$|\Psi\rangle = |0_a\rangle \left(\cos(\theta)\,|R\rangle + \sin(\theta)\,|S\rangle\right) \tag{9}$$

A fixed point operator, $\hat{F}(\theta)$ that marks the ancilla qubit a with the probability of the solution state $|S\rangle$ is defined for the orthogonal basis

$$\begin{aligned}\{&|0_a\rangle\,|R_1\rangle\,, |0_a\rangle\,|R_2\rangle\,, \cdots, |0_a\rangle\,|R_{N-M}\rangle\,,\\ &|0_a\rangle\,|S_1\rangle\,, |0_a\rangle\,|S_2\rangle\,, \cdots, |0_a\rangle\,|S_M\rangle\,,\\ &|1_a\rangle\,|R_1\rangle\,, |1_a\rangle\,|R_2\rangle\,, \cdots, |1_a\rangle\,|R_{N-M}\rangle\,,\\ &|1_a\rangle\,|S_1\rangle\,, |1_a\rangle\,|S_2\rangle\,, \cdots, |1_a\rangle\,|S_M\rangle\}.\end{aligned} \tag{10}$$

The column matrix of length $2N$ for the state $|\Psi\rangle$ in this basis would be given as

$$|\Psi\rangle = \begin{bmatrix} \cos(\theta)\frac{r_1}{|R|} \\ \vdots \\ \cos(\theta)\frac{r_{N-M}}{|R|} \\ \\ \sin(\theta)\frac{s_1}{|S|} \\ \vdots \\ \sin(\theta)\frac{s_M}{|S|} \\ \\ 0 \\ \vdots_{N \text{ times}} \\ 0 \end{bmatrix}. \tag{11}$$

The algorithm never leaves the four-dimensional invariant subspace defined by

$$\mathcal{V} = \text{span}\big\{|0_a\rangle\,|R\rangle\,,\ |0_a\rangle\,|S\rangle\,,\ |1_a\rangle\,|R\rangle\,,\ |1_a\rangle\,|S\rangle\big\}. \tag{12}$$

On this subspace $\mathcal{V}$, in the ordered basis $\big(|0_a\rangle\,|R\rangle\,, |0_a\rangle\,|S\rangle\,, |1_a\rangle\,|R\rangle\,, |1_a\rangle\,|S\rangle\big)$, the fixed-point operator is unitary. This is noted below:

$$\hat{F}(\theta)\big|_{\mathcal{V}} = \begin{pmatrix} -\cos 2\theta & 0 & 0 & -\sin 2\theta \\ -\sin 2\theta & 0 & 0 & \cos 2\theta \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \hat{F}^\dagger \hat{F} = I_4. \tag{13}$$

It is easy to verify that

$$\begin{aligned}\hat{F}(\theta)\ |0_a\rangle\,|R\rangle &= -\cos 2\theta\ |0_a\rangle\,|R\rangle - \sin 2\theta\ |0_a\rangle\,|S\rangle\,,\\ \hat{F}(\theta)\ |0_a\rangle\,|S\rangle &= |1_a\rangle\,|S\rangle\,,\end{aligned} \tag{14}$$

which fix the action of $\hat{F}(\theta)$ on the initial state

$$|\Psi\rangle = |0_a\rangle\,(\cos\theta\,|R\rangle + \sin\theta\,|S\rangle).$$

The resulting full-state action is stated in Eq. (15) below. Although $\hat{F}(\theta)$ depends on θ, its implementation as described in Fig. 2 uses only the oracle $\hat{O}$, the state-preparation operator $\hat{P}$ and its adjoint $\hat{P}^\dagger$, and basic gates, none of which requires the explicit value of θ or M. The circuit of Fig. 2 is a product of unitaries and therefore extends $\hat{F}(\theta)|_{\mathcal{V}}$ to a unitary on the full $2N$-dimensional space. Since the state is confined to $\mathcal{V}$ throughout, its action off $\mathcal{V}$ is immaterial. The circuit description of $\hat{F}$ is explained in the Appendix A.

Each application of $\hat{F}$ thus uses a single (ancilla-controlled) Oracle call. The quantum circuit for the operation $\hat{F}(\theta)$ using n qubits and an ancilla is shown in Fig. 2.

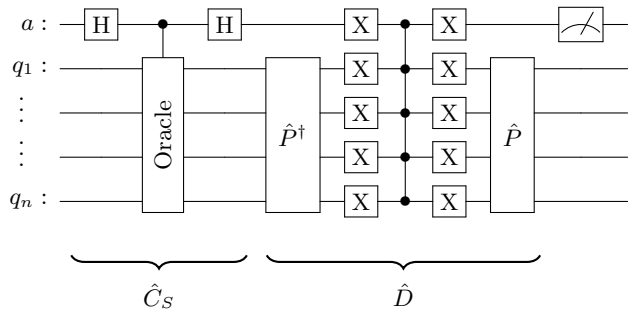


FIG. 2. Circuit realizing the fixed-point operator $\hat{F} = \hat{D}\,\hat{C}_S$, with the ancilla a drawn on the *top* wire. The first dashed block $\hat{C}_S = H_a\,(c\text{-}\hat{O})\,H_a$ writes the solution flag onto the ancilla; the second dashed block $\hat{D} = \hat{P}\,\hat{R}_0^{(a=0)}\,\hat{P}^\dagger$ performs the reflection about $|\psi\rangle$ conditioned on the ancilla being in $|0_a\rangle$. Here "Oracle" and $\hat{P}$ denote the circuits for the oracle $\hat{O}$ and the state-preparation operator $\hat{P}$. The full gate-level description is given in Appendix A.

The action of $\hat{F}(\theta)$ on the state $|\Psi\rangle$ is evaluated as

$$\hat{F}(\theta)|\Psi\rangle = -\cos\theta\,|0_a\rangle(\cos 2\theta|R\rangle + \sin 2\theta|S\rangle) + \sin\theta|1_a\rangle|S\rangle. \tag{15}$$

Here $\hat{F}(\theta)$ maps the state of the ancilla a to $|1_a\rangle$ simultaneously coupling the solution state $|S\rangle$ for the data qubits $q_1, q_2, \cdots, q_n$. This preserves the solution state's amplitude $\sin\theta$ from Eq. (2). Then, we measure the ancilla qubit a to check for convergence. If the state $|1_a\rangle$ is obtained (probability $\sin^2\theta$), we declare convergence to the solution state $|S\rangle$ for the data qubits. If the state $|0_a\rangle$ is obtained (probability $\cos^2\theta$), we get the state

$$|\Psi'\rangle = |0_a\rangle\,(\cos(2\theta)\,|R\rangle + \sin(2\theta)\,|S\rangle)\,, \tag{16}$$

where $|\Psi'\rangle$ is the state of the data qubits and ancilla a after measurement of a in state $|0_a\rangle$. This allows convergence to the state $|S\rangle$ after multiple measurements on ancilla a.

After an arbitrary iteration of the search, the ancilla qubit a is measured, and the state of the data qubits $q_1, q_2, \cdots, q_n$ and a is given as

$$|\Phi\rangle = |0_a\rangle\,|\phi\rangle = |0_a\rangle\,(\cos\phi\,|R\rangle + \sin\phi\,|S\rangle) \tag{17}$$

with an unknown angle ϕ, the solution amplitude is $\sin\phi$.

$\hat{F}(\theta)$ maps this amplitude to state $|1_a\rangle$ of ancilla a . The action of $\hat{F}(\theta)$ on $|\Phi\rangle$ is given as

$$\hat{F}\,|0_a\rangle\,|\phi\rangle = -\cos\phi\,|0_a\rangle\left(\cos 2\theta\,|R\rangle + \sin 2\theta\,|S\rangle\right) + \sin\phi\,|1_a\rangle\,|S\rangle \tag{18}$$

where $\sin\theta$ is the initial amplitude of solution state $|S\rangle$. The $\hat{F}(\theta)$ has successfully mapped the solution amplitude $\sin\phi$ of state $|\phi\rangle$ to the state $|1_a\rangle$ of ancilla a. The $|0_a\rangle$ branch retains the data state $\cos 2\theta\,|R\rangle + \sin 2\theta\,|S\rangle$, preserving the information of the initial state $|\psi\rangle$ given in Eq. (2) in the form of the angle 2θ. This allows repeated measurements of a to reach the solution state confidently. Measurement of the ancilla qubit a, changes the state $|\phi\rangle$ of the data qubits as follows.

Measurement

We measure the ancilla qubit a to be in state $|1_a\rangle$ with probability $\sin^2\phi$. If $|1_a\rangle$ is obtained, then the data qubits $q_1, q_2, \cdots, q_n$ collapse to the solution state $|S\rangle$, and we stop the search. After each failure in converging to the solution state (ancilla measured in $|0_a\rangle$), we get back the state $|\Phi_m\rangle$ after the measurement of ancilla a

$$|\Phi_m\rangle = |0_a\rangle\,|\phi_m\rangle = |0_a\rangle\,(\cos 2\theta\,|R\rangle + \sin 2\theta\,|S\rangle)\,. \tag{19}$$

The initial solution and non-solution state probabilities are preserved in the form of 2θ. Thus, the solution state $|S\rangle$ can be searched again in the next iteration. The final implementation combines the amplitude amplification and fixed-point methods above into a single search algorithm.

III. ALGORITHM AND ANALYSIS

In this section, we shall look at the final implementation of the fixed-point quantum search algorithm, and its success probability and query complexity calculations. Figure 3 contrasts the two behaviours: standard Grover search must halt at precisely the optimal iteration count, since its success probability oscillates and collapses on overshoot, whereas the ancilla-mediated scheme accumulates success across checkpoints and never regresses.

1. *Fixed point search algorithm*

The number of iterations ϵ_0 from Eq. (6) shall be maximum for $M = 1$ as

$$\epsilon_0 = \left\lfloor \frac{\pi}{4}\sqrt{N} - \frac{1}{2} \right\rfloor. \tag{20}$$

This represents the upper bound on the number of Grover iterations required for the search. After each failed check the state resets to angle 2θ [Eq. (16)]. We therefore index

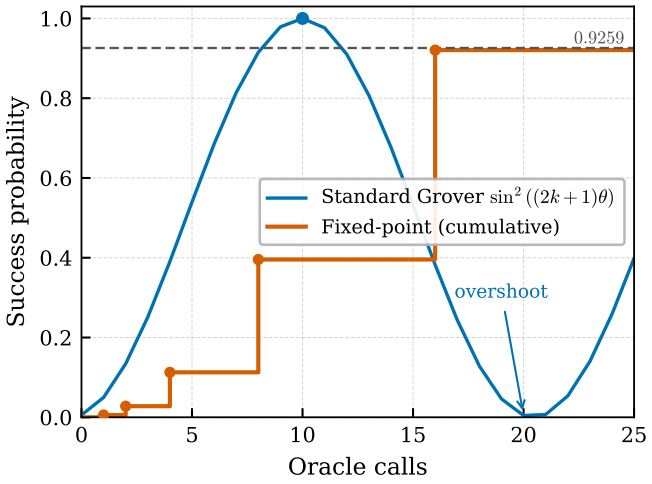


FIG. 3. The soufflé problem, at a representative $\theta = 0.075$. Standard Grover's single-shot success probability (blue) $\sin^2((2k+1)\theta)$ oscillates with the iteration count k; halting past the optimum (marked) collapses the probability ("overshoot"). The ancilla-mediated fixed-point scheme (orange) accumulates success across the doubling checkpoints $O_j = 2^j$; its cumulative success probability is non-decreasing by construction, staying above the analytical bound 0.9259 once reached.

the checks by $j = 0, 1, \ldots, r-1$ and apply the j-th check at the geometrically doubling angle

$$\varphi_j = 2^j\theta, \qquad j = 0, 1, \ldots, r-1, \tag{21}$$

where $\theta_{\min} = \arcsin\sqrt{1/N}$ is the smallest angle (worst case $M = 1$) and r is the smallest integer for which the doubling schedule reaches beyond the first amplitude peak, even in this worst case, i.e.,

$$2^{r-1}\theta_{\min} \geq \tfrac{\pi}{2} \implies r = \lceil \log_2(\pi\sqrt{N}) \rceil. \tag{22}$$

Starting from the reset angle 2θ, the angle φ_j is reached by

$$g_0 = 0, \qquad g_j = 2^{j-1} - 1 \quad (j \geq 1) \tag{23}$$

amplitude-amplification operations $\hat{G}$ (note $g_1 = 0$, since the reset already sits at angle 2θ; $g_0 = 0$, since the first check acts on the freshly prepared state at angle θ). The ordered schedule of Grover counts is $\mathcal{K} = \{g_0, g_1, \ldots, g_{r-1}\} = \{0, 0, 1, 3, 7, \ldots, 2^{r-2} - 1\}$. As in the Boyer–Brassard–Høyer–Tapp (BBHT) algorithm [9], the schedule doubles the coverage of the search space, but here in a deterministic fixed-point context; the first check at $j = 0$ gives a high chance of convergence when $M \sim N$.

These g_j amplifications carry the solution-state amplitude from $\sin 2\theta$ to $\sin(2^j\theta)$. Correspondingly, the solution-state amplitude set $\mathcal{S}_A$ is

$$\mathcal{S}_A = \{\sin\theta,\ \sin 2\theta,\ \sin 4\theta,\ \ldots,\ \sin(2^{r-1}\theta)\}. \tag{24}$$

With set $\mathcal{K}$ and solution amplitude $\mathcal{S}_A$, we present the Algorithm 1 for the full search operation. Fig. (4) represents the respective quantum circuit.

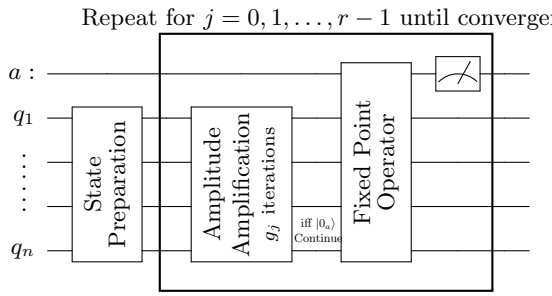


FIG. 4. Circuit of the fixed-point quantum search algorithm, with the ancilla a drawn on the top wire. At iteration j the data register is amplified by g_j Grover operations and then checked by the fixed-point operator $\hat{F}(\theta)$ (Fig. 2). If the ancilla is measured in $|1_a\rangle$ the data register has converged to the solution state $|S\rangle$, otherwise ($|0_a\rangle$) the data register has reset to angle 2θ and we continue to the next iteration. The convergence probability at the j-th check is $\sin^2(2^j\theta)$, $j \in \{0, 1, \ldots, r-1\}$. If the schedule completes without the ancilla ever being $|1_a\rangle$, we report that no solution was found, which is incorrect with probability at most $1 - 0.926 = 0.074$.

Algorithm 1 Ancilla-mediated fixed-point quantum search

Prepare $|\Psi\rangle \leftarrow |0_a\rangle \hat{P}\,|0\rangle^{\otimes n} = |0_a\rangle\,(\cos\theta\,|R\rangle + \sin\theta\,|S\rangle)$ ▷ angle θ
for $j = 0$ to $r-1$ **do**
  if $j \geq 1$ **then** ▷ amplify from the reset angle 2θ to $2^j\theta$
    for $g_j = 2^{j-1} - 1$ iterations **do**
      $|\Psi\rangle \leftarrow \hat{G}\,|\Psi\rangle$
    end for
  end if
  $|\Psi\rangle \leftarrow \hat{F}(\theta)\,|\Psi\rangle$ ▷ apply the fixed-point check
  Measure ancilla a ▷ success probability $\sin^2(2^j\theta)$
  if a is in $|1\rangle$ **then**
    Search complete; the data qubits are in the solution state $|S\rangle$.
    stop
  end if
  ▷ else the state has reset to $|0_a\rangle\,(\cos 2\theta\,|R\rangle + \sin 2\theta\,|S\rangle)$
end for
return "no solution found" ▷ bounded error: incorrect w.p. ≤ 0.074

2. Probability calculations and Analysis

In this subsection, we examine the probability calculations and the worst-case minimum probability of success of the algorithm. Algorithm 1 gives the probability p_s of converging to the solution state at the j-th check as

$$p_s = \sin^2(2^j\theta) \tag{25}$$

where g_j is the number of amplitude-amplification operations $\hat{G}$ applied before the j-th check [Eq. (23)]. The probability of failing to collapse to $|S\rangle$ at that check is $\cos^2(2^j\theta)$. By Eqs. (5), (6) and (22), there exists an index $c \leq r-2$ (so that both checks c and $c+1$ lie within the schedule) satisfying

$$\frac{\pi}{4} \leq 2^c\theta < \frac{\pi}{2}. \tag{26}$$

The probability $p_{c,1}$ of converging within the two consecutive checks at indices c and $c+1$ is

$$p_{c,1} = \sin^2(2^c\theta) + \sin^2(2^{c+1}\theta)\,\cos^2(2^c\theta). \tag{27}$$

Let $x = 2^c\theta$; by Eq. (26) the admissible range is $x \in [\frac{\pi}{4}, \frac{\pi}{2})$, and since $2^{c+1}\theta = 2x$ this is the single-variable function

$$p_{c,1}(x) = \sin^2 x + \sin^2(2x)\cos^2 x = 1 - \cos^2 x\,\cos^2(2x), \tag{28}$$

the second form following from $\sin^2\alpha = 1 - \cos^2\alpha$. On $[\frac{\pi}{4}, \frac{\pi}{2})$ the product $\cos^2 x\,\cos^2(2x)$ attains its maximum 0.07407 at $x = 1.15026$, so $p_{c,1}$ attains its minimum $1 - 0.07407 = 0.92593$ there, with $p_{c,1} = 1$ at the endpoint $x = \frac{\pi}{4}$. Because the measurement angles $2^j\theta$ double from θ, exactly one of them lies in $[\frac{\pi}{4}, \frac{\pi}{2})$ for every $\theta \in (0, \frac{\pi}{2})$, so a valid index c always exists (the boundary case $\theta = \frac{\pi}{2}$, i.e. $M = N$, converges already at the first check).

The probability of first converging at index j (all earlier checks having failed and reset) is

$$P_{S_j} = \sin^2(2^j\theta)\prod_{m=0}^{j-1}\cos^2(2^m\theta), \tag{29}$$

and the total probability of converging by index j telescopes to

$$P_j = \sum_{l=0}^{j}\sin^2(2^l\theta)\prod_{m=0}^{l-1}\cos^2(2^m\theta) = 1 - \prod_{m=0}^{j}\cos^2(2^m\theta), \tag{30}$$

where j is the last index in the schedule. Since every factor $\cos^2(2^m\theta) \leq 1$,

$$P_j \;\geq\; 1 - \cos^2(2^c\theta)\,\cos^2(2^{c+1}\theta) \;=\; p_{c,1} \;\geq\; 0.9259, \tag{31}$$

Figure 5 shows these probability values for $\mathcal{O}(\sqrt{N/M})$ iterations in relation to M for $N = 10^{10}$. Figure 6 shows how the cumulative success probability climbs across the doubling checkpoints for representative problem sizes.

3. Query complexity

In this subsection, we shall examine the query complexity required to achieve the stated success probability of 0.926.

Each application of $\hat{G}$ and each application of $\hat{F}$ is one oracle call, so the j-th check costs

$$o_j = 1 + g_j = \begin{cases} 1, & j = 0, \\ 2^{j-1}, & j \geq 1. \end{cases} \tag{32}$$

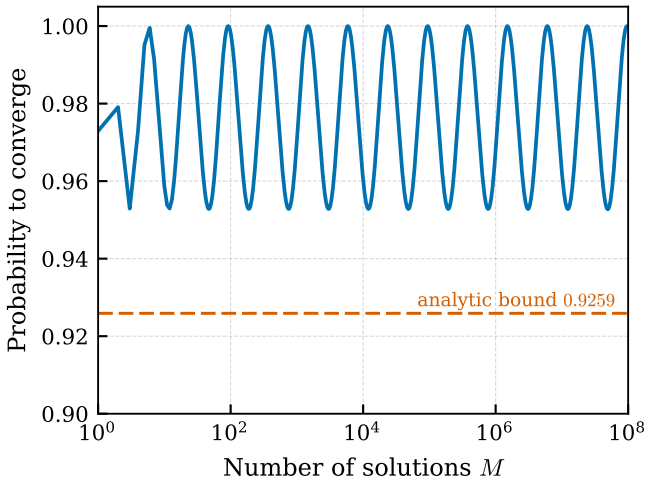


FIG. 5. Probability of converging to a solution state within $\mathcal{O}(\sqrt{N/M})$ oracle calls versus M, for $N = 10^{10}$. The dashed line marks the analytical lower bound 0.9259; over the plotted range the numerical minimum is ≈ 0.95, comfortably above it.

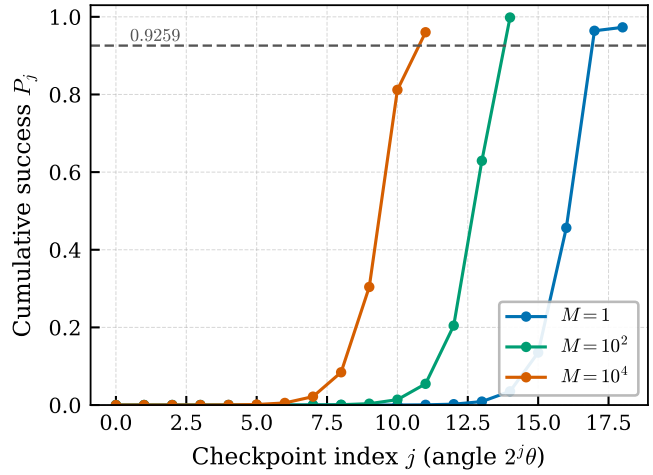


FIG. 6. Cumulative success probability P_j versus checkpoint index j (measurement angle $2^j\theta$) for $M = 1$, 10^2, 10^4 at $N = 10^{10}$. Larger M (larger θ) converges in fewer doublings; each curve crosses the analytical bound 0.9259 (dashed).

The cumulative number of oracle calls to reach the j-th check is therefore

$$O_j = \sum_{l=0}^{j} o_l = 2^j. \tag{33}$$

By Eq. (26), the two-index guarantee of at least 92.6% is met at the check with index $c+1$, reaching it costs

$$O_T = O_{c+1} = 2^{c+1} < \pi\sqrt{\tfrac{N}{M}}, \tag{34}$$

where the bound uses $2^c\theta < \frac{\pi}{2}$ and $\theta \simeq \sqrt{M/N}$. Hence the worst-case complexity of converging to $|S\rangle$ with probability at least 92.6% is $\mathcal{O}(\pi\sqrt{N/M})$. We have plotted

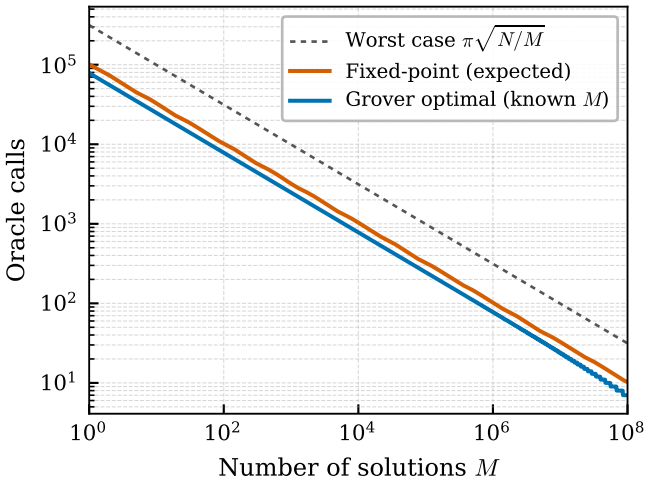


FIG. 7. Oracle calls versus M for $N = 10^{10}$: Grover's optimal count assuming M is known, the expected cost of the ancilla-mediated fixed-point algorithm, and the worst-case guarantee $\pi\sqrt{N/M}$. The expected cost exceeds the known-M optimum by a factor ≈ 1.33, while the worst case stays within $\pi\sqrt{N/M}$.

the expected oracle calls $\mathbb{E}(O_j)$ before converging on index j from Eq. (30) as

$$\mathbb{E}(O_j) = \sum_{l=0}^{j} 2^l \sin^2(2^l\theta) \prod_{m=0}^{l-1} \cos^2(2^m\theta) + 2^j \prod_{m=0}^{j} \cos^2(2^m\theta) \tag{35}$$

in Fig (7). $\mathbb{E}(O_j)$ comes out to be approximately $1.33 \times \epsilon$ as

$$\mathbb{E}(O_j) \sim 1.33\,\epsilon, \tag{36}$$

for a small θ. Further, using the value of ϵ from Eq. 6, we get $\mathbb{E}(O_j)$ as

$$\mathbb{E}(O_j) \sim \sqrt{\frac{N}{M}}. \tag{37}$$

Hence, we show that the algorithm can theoretically reach the success probability higher than 0.926 in less than $\mathcal{O}(\pi\sqrt{N/M})$ oracle calls, thus maintaining the quadratic speedup. We further numerically demonstrate that the expected number of oracle calls to reach a success probability of 0.95 is approximately $\mathcal{O}(\sqrt{N/M})$ for a dataset of size $N = 10^{10}$.

IV. DISCUSSION AND CONCLUSION

In this section, we summarize our work, compare it with other existing quantum search algorithms, and provide a brief conclusion. Unlike standard amplitude amplification, where over-rotation leads to a total loss of the solution state's overlap, our ancilla-mediated approach

allows for repeated checkpoint measurements. If the ancilla is measured as $|0_a\rangle$, the system remains in a coherent superposition that preserves the initial state information, allowing the search to continue from a known state rather than restarting from scratch. The algorithm can be modified to serve as an oblivious fixed-point algorithm, marking solution states using calculation operations introduced by Berry *et al.* in 2014 as the oblivious amplitude amplification method [26].

Our work has a distinct advantage over existing quantum search methods with an expected query complexity of approximately $\mathcal{O}(\sqrt{N/M})$. Unlike prior works, our algorithm does not utilize phase gates and is geometrically similar to Grover's algorithm reflection-based approach.

A structural feature of this reflection-based architecture is that each fixed-point check uses a single ancilla-controlled oracle call together with one oracle-free reflection about $|\psi\rangle$ built from $\hat{P}, \hat{P}^\dagger$ (Appendix A). This matches the one-oracle-per-iteration cost of the generalized Grover iterate underlying the phase-matching methods [16, 17], in which the target reflection consumes one oracle query while the reflection about the initial state is oracle-free; the distinction of our construction is therefore not a reduced query count but its phase-free, real-plane realization using a single measurable ancilla. We show a significant complexity advantage over existing work of [17] (complexity $\mathcal{O}(5.643\sqrt{N/M})$) by achieving a success probability of 92.6% in $\mathcal{O}(\pi\sqrt{N/M})$ worst case queries. We note that the phase-matching methods of Yoder *et al.* [16] and Li *et al.* [17] deliver a tunable success probability $1-\delta^2$, so their query constants are quoted at a fixed δ; our method instead fixes a $\geq 92.6\%$ guarantee with a simpler, phase-free circuit.

Our work has a deterministic iteration schedule with a distinct upper bound for search termination, which differentiates it from existing randomized approaches.

Table (I) gives the mechanism, convergence, query complexity and success probability of existing search algorithms and ours. Further, our work provides clear quantum circuits for easy implementation of the algorithm for future use.

The fixed-point quantum search paradigm achieves robust convergence; by decoupling the success probability from the precise knowledge of M, we have circumvented the soufflé problem. The algorithm achieves a stable success rate of $\geq 92.6\%$ while ensuring that subsequent iterations do not degrade solution quality, a common failure mode in unstructured search. The fixed-point search has a query complexity of $\mathcal{O}(\sqrt{N/M})$, similar to Grover's search, without requiring the optimal number of iterations. This solves a major problem with the practical applications of Grover's search algorithm for searching an unknown dataset while maintaining its core architecture. This makes this method geometrically distinct from prior phase-based fixed-point approaches while maintaining similar or better time-complexity performance. In the worst case, with an unknown number of solution states M, the proposed algorithm requires at most $\pi\sqrt{N/M}$ oracle calls. The expected number of oracle calls to converge to a solution is $\mathcal{O}(\sqrt{N/M})$ with a probability of at least 92.6%. This architecture attempts to solve the search problem inspired by Grover's algorithm, achieving $\mathcal{O}(\sqrt{N/M})$ complexity when M is unknown. Future work could explore integrating the fixed-point operator into adaptive search frameworks for quadratic binary optimization, where the number of solutions changes dynamically and is inherently unknown [27].

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TABLE I. A comparison of quantum search algorithms

Algorithm	**Mechanism**	**Convergence**	**Query Complexity**	**Success Prob.**
Grover F.P. [14]	$\pi/3$ phase-matching F.P.	Deterministic	$\ln(1/\delta)/\lambda$ [b]	$1-\delta^2$ [c]
Boyer et al. [9, 17]	Trial-and-error	Randomized	$4/\sqrt{\lambda_0}$ [a]	$\geq 1/4$ per trial
Yoder et al. [16]	Phase-matching F.P.	Deterministic	$\ln(2/\delta)/\sqrt{\lambda}$ [b]	$1-\delta^2$ [c]
Li et al. [17]	Trial-and-error + phase-matching hybrid	Deterministic	$5.643/\sqrt{\lambda_0}$ [a]	$1-\delta^2$ [c]
This work	Ancilla-mediated real-plane F.P.	Deterministic	$\pi/\sqrt{\lambda_0}$ [a,d]	$\geq 92.6\%$

[a] λ_0 denotes the fraction M/N; see Eq. (6).
[b] Requires a known lower bound λ for the fraction M/N [16]
[c] δ is the optimal parameter to minimize the expected query complexity [16, 17]
[d] Expected number of oracle calls are approximately $\mathcal{O}(\frac{1}{\sqrt{\lambda_0}})$.

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Appendix A: Gate-level realization of $\hat{F}(\theta)$

This appendix derives the fixed-point operator $\hat{F}(\theta)$ directly from the circuit of Fig. 2 and verifies, step by step, that the circuit reproduces the 4×4 unitary of Eq. (13) and the basis-state actions of Eq. (14).

1. Decomposition into sub-operators

In operator order (the rightmost factor acting first on the state), the circuit of Fig. 2 factorises into two sub-operators:

$$\hat{F}(\theta) = \underbrace{\left(\hat{P}\,\hat{R}_0^{(a=0)}\,\hat{P}^\dagger\right)}_{\hat{D}}\;\underbrace{\left(H_a\,\text{c-}\hat{O}\,H_a\right)}_{\hat{C}_S}. \tag{A1}$$

When the combined state $|0_a\rangle\,|\psi\rangle$ enters the circuit, it first passes through $\hat{C}_S$ (the *ancilla-flag* sub-operator) and then through $\hat{D}$ (the *conditional-reflection* sub-operator). We now describe the individual gates that compose each of these two blocks and explain the physical role of every element.

2. Circuit elements and their roles

The circuit employs the following gates, listed in the order they appear from left to right in Fig. 2:

a. Hadamard on the ancilla (H_a). The first H_a maps $|0_a\rangle \to |+_a\rangle = (|0_a\rangle + |1_a\rangle)/\sqrt{2}$, creating a superposition of the ancilla states. This enables subsequent controlled-oracle to imprint the solution information onto the ancilla via *phase kickback* [1].

b. Ancilla-controlled oracle (c-$\hat{O}$). The phase oracle $\hat{O}$ is applied to the n data qubits, *controlled on the ancilla being* $|1\rangle$. On the $|1_a\rangle$ branch of the superposition, $\hat{O}$ flips the phase of solution states ($|S\rangle \to -|S\rangle$) while leaving non-solution states ($|R\rangle$) unchanged. On the $|0_a\rangle$ branch, nothing happens.

c. Second Hadamard on the ancilla (H_a). For a non-solution state $|R\rangle$, the two branches carry the same phase, so H_a returns the ancilla to $|0_a\rangle$. For a solution state $|S\rangle$, the $|1_a\rangle$ branch has acquired a -1 phase from the oracle, so H_a maps the ancilla to $|1_a\rangle$.

d. Inverse state preparation ($\hat{P}^\dagger$). Acting on the data qubits only, $\hat{P}^\dagger$ maps the initial superposition $|\psi\rangle = \cos\theta\,|R\rangle + \sin\theta\,|S\rangle$ back to $|0\rangle^{\otimes n}$. Together with $\hat{P}$ on the other side, it conjugates the computational-basis reflection $\hat{R}_0^{(a=0)}$ into a reflection about $|\psi\rangle$.

e. Conditional multi-controlled-Z reflection ($\hat{R}_0^{(a=0)}$). This is the central multi-qubit gate. In the computational basis it is defined as

$$\hat{R}_0^{(a=0)} = I - 2\,|0_a\rangle\langle 0_a| \otimes |0\rangle^{\otimes n}\langle 0|^{\otimes n}, \tag{A2}$$

which places a -1 phase on the joint state $|0_a\rangle\,|0\rangle^{\otimes n}$ and acts as the identity on every other basis state. In particular, it acts as the identity whenever the ancilla is $|1_a\rangle$.

The standard decomposition of this gate is: (i) apply X to each of the n data qubits, mapping $|0\rangle^{\otimes n} \to |1\rangle^{\otimes n}$; (ii) apply a multi-controlled-Z gate with the n data qubits as $|1\rangle$-controls and the ancilla as a $|0\rangle$-control. In other words, the ancilla must be $|0\rangle$ for the phase to be applied, which can be realised by sandwiching the ancilla's control wire with X gates in a standard multi-controlled-Z; (iii) apply X to each data qubit again to undo step (i). This is the X–MCZ–X structure depicted in the right half of Fig. 2.

f. State preparation ($\hat{P}$). $\hat{P}$ re-prepares the data-register superposition $|\psi\rangle = \hat{P}\,|0\rangle^{\otimes n}$, completing the conjugation $\hat{D} = \hat{P}\,\hat{R}_0^{(a=0)}\,\hat{P}^\dagger$.

g. Ancilla measurement. The measurement at the end of the circuit reads the ancilla in the computational basis. Outcome $|1_a\rangle$ signals that the data register has converged to a solution state. Outcome $|0_a\rangle$ signals otherwise, and the data register is recycled into the next round.

3. Step 1: Ancilla flag ($\hat{C}_S$)

We now verify the action of $\hat{C}_S = H_a\,(\text{c-}\hat{O})\,H_a$ on the two basis states $|0_a\rangle\,|R\rangle$ and $|0_a\rangle\,|S\rangle$ that span the initial subspace.

a. Action on $|0_a\rangle\,|R\rangle$ (non-solution).

$$\begin{aligned}
|0_a\rangle\,|R\rangle &\xrightarrow{H_a} \tfrac{1}{\sqrt{2}}\big(|0_a\rangle + |1_a\rangle\big)\,|R\rangle \\
&\xrightarrow{\text{c-}\hat{O}} \tfrac{1}{\sqrt{2}}\big(|0_a\rangle\,|R\rangle + |1_a\rangle\,\underbrace{\hat{O}\,|R\rangle}_{=|R\rangle}\big) = \tfrac{1}{\sqrt{2}}\big(|0_a\rangle + |1_a\rangle\big)\,|R\rangle \\
&\xrightarrow{H_a} |0_a\rangle\,|R\rangle\,.
\end{aligned}$$

Since $|R\rangle$ is a non-solution state, the oracle leaves it unchanged in both branches. The second Hadamard returns the ancilla to $|0_a\rangle$. In summary:

$$\hat{C}_S\,|0_a\rangle\,|R\rangle = |0_a\rangle\,|R\rangle\,. \tag{A3}$$

b. Action on $|0_a\rangle\,|S\rangle$

$$\begin{aligned}
|0_a\rangle\,|S\rangle &\xrightarrow{H_a} \tfrac{1}{\sqrt{2}}\big(|0_a\rangle + |1_a\rangle\big)\,|S\rangle \\
&\xrightarrow{\text{c-}\hat{O}} \tfrac{1}{\sqrt{2}}\big(|0_a\rangle\,|S\rangle + |1_a\rangle\,\underbrace{\hat{O}\,|S\rangle}_{=-|S\rangle}\big) \\
&= \tfrac{1}{\sqrt{2}}\big(|0_a\rangle - |1_a\rangle\big)\,|S\rangle = |-_a\rangle\,|S\rangle \\
&\xrightarrow{H_a} |1_a\rangle\,|S\rangle\,.
\end{aligned}$$

The oracle flips the phase of $|S\rangle$ only in the $|1_a\rangle$ branch. The resulting relative minus sign converts $|+_a\rangle \to |-_a\rangle$, and the second Hadamard maps $|-_a\rangle \to |1_a\rangle$. This is the phase-kickback mechanism: the oracle's phase information has been transferred to a bit flip on the ancilla. In summary:

$$\hat{C}_S\,|0_a\rangle\,|S\rangle = |1_a\rangle\,|S\rangle\,. \tag{A4}$$

$\hat{C}_S$ routes solution states to $|1_a\rangle$ and non-solution states to $|0_a\rangle$, without collapsing any superposition in the data register.

4. Step 2: Conditional reflection ($\hat{D}$)

The conditional-reflection sub-operator is

$$\hat{D} = \hat{P}\,\hat{R}_0^{(a=0)}\,\hat{P}^\dagger. \tag{A5}$$

We first work out its structure and then compute its action on the states produced by $\hat{C}_S$.

a. Structure. Conjugating $\hat{R}_0^{(a=0)}$ by $\hat{P}$ (which acts only on the data register) gives

$$\hat{D} = |0_a\rangle\langle 0_a| \otimes \big(I - 2\,|\psi\rangle\langle\psi|\big) + |1_a\rangle\langle 1_a| \otimes I, \tag{A6}$$

because $\hat{P}\,|0\rangle^{\otimes n}\langle 0|^{\otimes n}\,\hat{P}^\dagger = |\psi\rangle\langle\psi|$ in the $|0_a\rangle$ sector, and $\hat{R}_0^{(a=0)}$ acts as the identity whenever the ancilla is $|1_a\rangle$.

Equation (A6) shows that $\hat{D}$ is an ancilla-controlled operation: when the ancilla is $|0_a\rangle$, the data register undergoes a reflection about $|\psi\rangle$; when the ancilla is $|1_a\rangle$, the data register is untouched.

b. Reflection on $|R\rangle$. Using $|\psi\rangle = \cos\theta\,|R\rangle + \sin\theta\,|S\rangle$ and hence $\langle\psi|R\rangle = \cos\theta$:

$$\begin{aligned}\big(I - 2\,|\psi\rangle\langle\psi|\big)\,|R\rangle &= |R\rangle - 2\cos\theta\;|\psi\rangle \\ &= |R\rangle - 2\cos\theta\big(\cos\theta\,|R\rangle + \sin\theta\,|S\rangle\big) \\ &= (1 - 2\cos^2\theta)\,|R\rangle - 2\sin\theta\cos\theta\,|S\rangle \\ &= -\cos 2\theta\;|R\rangle - \sin 2\theta\;|S\rangle\,, \qquad \text{(A7)}\end{aligned}$$

where we used the double-angle identities $\cos 2\theta = 2\cos^2\theta - 1$ and $\sin 2\theta = 2\sin\theta\cos\theta$.

c. Reflection on $|S\rangle$. Similarly, $\langle\psi|S\rangle = \sin\theta$, so

$$\begin{aligned}\big(I - 2\,|\psi\rangle\langle\psi|\big)\,|S\rangle &= |S\rangle - 2\sin\theta\;|\psi\rangle \\ &= -\sin 2\theta\;|R\rangle + \cos 2\theta\;|S\rangle\,. \qquad \text{(A8)}\end{aligned}$$

5. Composition: Full action of $\hat{F}(\theta) = \hat{D}\,\hat{C}_S$

We now compose the two sub-operators on each basis state.

a. On $|0_a\rangle\,|R\rangle$. From Step 1, $\hat{C}_S\,|0_a\rangle\,|R\rangle = |0_a\rangle\,|R\rangle$ with the ancilla remaining as $|0_a\rangle$. The ancilla being $|0_a\rangle$ activates the reflection in $\hat{D}$ (Eq. (A6)), which by Eq. (A7) gives

$$\hat{F}(\theta)\,|0_a\rangle\,|R\rangle = -\cos 2\theta\;|0_a\rangle\,|R\rangle - \sin 2\theta\;|0_a\rangle\,|S\rangle\,. \quad \text{(A9)}$$

b. On $|0_a\rangle\,|S\rangle$. From Step 1, $\hat{C}_S\,|0_a\rangle\,|S\rangle = |1_a\rangle\,|S\rangle$ where the ancilla has flipped to $|1_a\rangle$. Since the ancilla is now $|1_a\rangle$, $\hat{D}$ acts as the identity (Eq. (A6)), leaving

$$\hat{F}(\theta)\,|0_a\rangle\,|S\rangle = |1_a\rangle\,|S\rangle\,. \qquad \text{(A10)}$$

Both results agree with Eq. (14) and furnish the first two columns of the matrix in Eq. (13) (the action on $|0_a\rangle\,|R\rangle$ and $|0_a\rangle\,|S\rangle$, respectively). Repeating the same computation on the remaining basis states gives $\hat{F}(\theta)\,|1_a\rangle\,|R\rangle = |1_a\rangle\,|R\rangle$ and $\hat{F}(\theta)\,|1_a\rangle\,|S\rangle = -\sin 2\theta\;|0_a\rangle\,|R\rangle + \cos 2\theta\;|0_a\rangle\,|S\rangle$, which reproduce the third and fourth columns and confirm that $\mathcal{V}$ is invariant under $\hat{F}(\theta)$. The circuit therefore realizes exactly the unitary of Eq. (13) on $\mathcal{V}$; the $|1_a\rangle$ sector is never populated by the algorithm and plays no role in the search.

6. Gate count and oracle cost

Each application of $\hat{F}(\theta)$ consists of:

1. Two single-qubit Hadamards on the ancilla in $\hat{C}_S$.
2. One ancilla-controlled oracle call c-$\hat{O}$ in $\hat{C}_S$.
3. One application each of $\hat{P}^\dagger$ and $\hat{P}$ in $\hat{D}$.
4. Two layers of $X^{\otimes n}$ on the data qubits, together with two X gates on the ancilla that convert it into a $|0\rangle$-control (in $\hat{R}_0^{(a=0)}$).
5. One $(n{+}1)$-qubit multi-controlled-Z gate $\hat{R}_0^{(a=0)}$, whose standard decomposition into one- and two-qubit gates uses $\mathcal{O}(n)$ elementary gates and does not depend on θ.

In total, each application of $\hat{F}(\theta)$ uses exactly *one* oracle call, together with $\mathcal{O}(n)$ single-qubit gates and one multi-controlled-Z gate. The single oracle call originates from the ancilla-controlled oracle c-$\hat{O}$ in $\hat{C}_S$; the reflection $\hat{D}$ about $|\psi\rangle$ is built solely from $\hat{P}, \hat{P}^\dagger$ and basic gates and makes no oracle query. This one-query-per-check cost coincides with that of the generalized Grover iterate underlying the phase-matching methods [16, 17], whose target reflection consumes one oracle query while the reflection about the initial state is oracle-free. The distinguishing feature of the present construction is therefore not a reduced oracle count but its phase-free, real-plane realization: the fixed-point check is assembled entirely from Grover-type reflections and a single measurable ancilla, with no tunable phase gates.